

Applications of Lebesgue Measurability

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Introduction

- ▶ Lebesgue measure is a modern method of measuring the size of sets.
- ▶ Lebesgue measurability tells us which sets can be assigned a consistent measure.
- ▶ It is more general than elementary length and is a foundation of modern real analysis.
- ▶ Main applications: integration, probability, convergence theorems, differential equations and mathematical physics.

Lebesgue Measure

- ▶ For an interval $E = (a, b)$, the Lebesgue measure is

$$m(E) = b - a.$$

- ▶ Example: $E = (2, 7)$. Therefore

$$m(E) = 7 - 2 = 5.$$

- ▶ A single point has measure zero:

$$m(\{x\}) = 0.$$

Lebesgue Outer Measure

For a set $E \subseteq \mathbb{R}$, Lebesgue outer measure is defined by covering E with countably many intervals:

$$m^*(E) = \inf \left\{ \sum_{n=1}^{\infty} |I_n| : E \subseteq \bigcup_{n=1}^{\infty} I_n \right\}.$$

- ▶ Here $|I_n|$ denotes the length of the interval I_n .
- ▶ Outer measure is the starting point for defining Lebesgue measurable sets.

Lebesgue Measurability

A set E is Lebesgue measurable if, for every $A \subseteq \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A - E).$$

This is called Carathéodory's criterion.

- Intervals, open sets, closed sets and countable sets are Lebesgue measurable.

Important Properties

- ▶ Non-negativity:

$$m(E) \geq 0.$$

- ▶ Empty set:

$$m(\emptyset) = 0.$$

- ▶ Monotonicity:

$$E \subseteq F \implies m(E) \leq m(F).$$

- ▶ Countable additivity: for pairwise disjoint measurable sets E_n ,

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n).$$

Application: Lebesgue Integration

- ▶ Lebesgue measurability allows us to define the integral of measurable functions:

$$\int f \, dm.$$

- ▶ Example: $f(x) = x$ on $E = [0, 1]$.

$$\int_0^1 x \, dm = \int_0^1 x \, dx = \frac{1}{2}.$$

- ▶ This method can handle many functions and sets that are difficult for elementary Riemann integration.

Application: Probability

- ▶ A probability space contains measurable events.
- ▶ For a uniform random variable X on $[0, 1]$, for measurable $E \subseteq [0, 1]$,

$$P(X \in E) = m(E).$$

- ▶ Example: $E = [0, 1/4]$. Then

$$P(X \in E) = \frac{1}{4}.$$

- ▶ Thus measure theory provides a mathematical foundation for continuous probability.

Application: Differential Equations

- ▶ Modern differential equations often use measurable and integrable functions.
- ▶ In weak-solution methods, a solution may not have classical derivatives everywhere.
- ▶ Lebesgue integration is used to formulate integral identities and weak derivatives.
- ▶ This is particularly important in the study of partial differential equations.

Application: Mathematical Physics

- ▶ Lebesgue integration is used in continuous mathematical models.
- ▶ Applications include probability distributions, quantum mechanics and statistical mechanics.
- ▶ Measurable domains and functions allow physical quantities to be integrated rigorously.
- ▶ Measure theory also underlies many function spaces used in mathematical physics.

Solved Mathematical Example

Let

$$E = (0, 1) \cup (2, 5).$$

The intervals are disjoint and measurable.

Therefore, by additivity,

$$\begin{aligned} m(E) &= m((0, 1)) + m((2, 5)) \\ &= (1 - 0) + (5 - 2) \\ &= 1 + 3 \\ &= 4. \end{aligned}$$

Hence the Lebesgue measure of E is 4.

Conclusion

- ▶ Lebesgue measurability is a fundamental concept in modern analysis.
- ▶ It connects measure and integration and supports important convergence results.
- ▶ Its applications extend to probability, differential equations and mathematical physics.

Measure $\longrightarrow$ Integration $\longrightarrow$ Probability $\longrightarrow$ Modern Analysis